

Secure Face Registration and Verification Framework

Final Project Report

A local system that registers a person's face once, then later confirms it is really them, and a real live person, before granting access.

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Report Date: August 6, 2026

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Executive Summary

This project builds a system that can recognize a person's face to let them log in or prove their identity -- the same idea behind Face ID on a phone or a fingerprint scanner at an office door, but built entirely from free, open tools running on a normal laptop, with no data ever leaving the machine.

The problem it solves: many face-recognition systems can be tricked by simply holding up a photo or playing a video of the real person. This project adds two extra checks beyond just matching a face: a check that the photo itself is clear and usable, and a check that a real, live person is actually in front of the camera right now, not a picture or a recording.

Does it work: yes, for the core flow. A person can register their face once, and later have it recognized reliably, with the system correctly telling the right person apart from someone else in internal testing. Real weaknesses were also found through honest testing rather than assumed away -- most notably, one of the three "is this a real person" checks can be fooled by a specific kind of screen replay, and fairness across different groups of people has not yet been fully addressed. Those are stated openly in this report, not hidden.

Honest current state: the underlying engineering is solid and has been tested with real data throughout, real problems were found and fixed rather than assumed away, and every known weakness is written down in this report. What remains is a short, known list, not a mystery: a recorded demonstration video, and one final confirmation pass on the most recent fixes.

The Problem

Face verification means using a photograph of someone's face to prove they are who they say they are, similar to how a password proves it on a website, except the "password" is your face.

The reason this is harder than it sounds is that a face-matching system alone can be tricked. If the system's only job is checking whether a face matches the one on file, someone could hold up a printed photo of the real person, play a video of them, or show a photo on a phone screen, and the system might accept it, because a photo of a face still looks like a face.

This is why the project needed a second capability on top of matching, called liveness detection: a way to check that a real, living person is actually in front of the camera right now, not a photo, a video, or a screen. It also needed a third capability, called image quality assessment, to make sure the photo being checked is clear and usable in the first place, since a blurry or badly lit photo can cause the wrong decision either way.

The Journey, Phase by Phase

This section walks through the project in the order it actually happened, including the real problems hit along the way and how each was actually fixed -- not just a list of finished features. Figure 1, next page, shows the seven phases in order.

Figure 1 -- Project Timeline



Phase 1: Planning and Reviewing the Original Brief

What we were trying to do: before writing a single line of code, read the original project brief carefully and honestly point out anything missing in the plan itself, rather than assume it was ready to build from.

What we actually built: a real 15-day day-by-day plan, a choice of tools (Python, a face-detection library called MediaPipe, a face-matching library called DeepFace, and a simple local database called SQLite -- all free and able to run on a normal laptop), and a daily log so every decision was written down at the time, not remembered later from memory.

The real problem we found: the brief did not say which sets of face photos, called datasets, should be used to test the system, did not explain how to pick the exact sensitivity numbers ("thresholds") the checks should use, and its definition of success was vague -- something like "acceptable performance" instead of an actual number to aim for.

How we fixed it: rather than guess, we picked specific, named datasets for specific jobs, and committed to measuring real numbers from real test photos to set every threshold, instead of picking numbers that simply felt reasonable.

What changed as a result: every number used later in this project -- thresholds, pass rates, error rates -- can be traced back to a real measurement, not a guess.

Phase 2: Building the Basic Photo-Quality Checks

What we were trying to do: teach the system to reject a bad photo before doing anything else with it -- too dark, too blurry, face turned too far sideways, more than one face in frame, or part of the face covered.

What we actually built: six separate checks, each looking at one specific thing: overall brightness, sharpness (called "blur" -- how out-of-focus the image is), head angle (how far the face is turned or tilted, called "pose"), whether the face is centered and takes up enough of the frame, whether exactly one face is visible, and whether part of the face looks covered or hidden.

We did not guess the cutoff numbers for any of these checks. Real photos were taken of a real person -- some good, some deliberately bad -- and the actual numbers each type of photo produced were measured, with each threshold placed in the real gap between "clearly good" and "clearly bad" photos.

The real bug we found: the head-angle check uses a technique called solvePnP, which works out the angle of a face in 3D space by matching known points on a face -- like the nose tip and eye corners -- against a generic 3D model of a face. When this was first tested against real video of a real face turning left and right, the system reported faces as upside-down that were clearly right-side-up on screen.

Separately, in some frames the same face angle produced two different, opposite-looking answers -- for example, reporting a face as turned left when it was actually turned right.

How we found it: this was caught by testing against real recorded footage of a person moving their head, not by reasoning about the code -- the mismatch between what the screen showed and what the system reported was immediately obvious once real video was used instead of a handful of still photos.

How we fixed it: the root cause was a mismatch in coordinate conventions -- the 3D reference points and the coordinate system the underlying math library expected were not lined up the same way, so the numbers coming out were technically correct answers to the wrong question. This was fixed by correcting the coordinate setup itself, not by adding a rule to flip the answer when it looked wrong, since patching the symptom would not have stopped the same bug from causing different, harder-to-spot errors elsewhere.

What changed as a result: head-angle detection became accurate and consistent across real test footage, and this fix mattered beyond just this one check -- the same angle numbers are reused for the head-turn liveness challenge in Phase 3.

Phase 3: Building the Three-Layer "Is This a Real Person" System

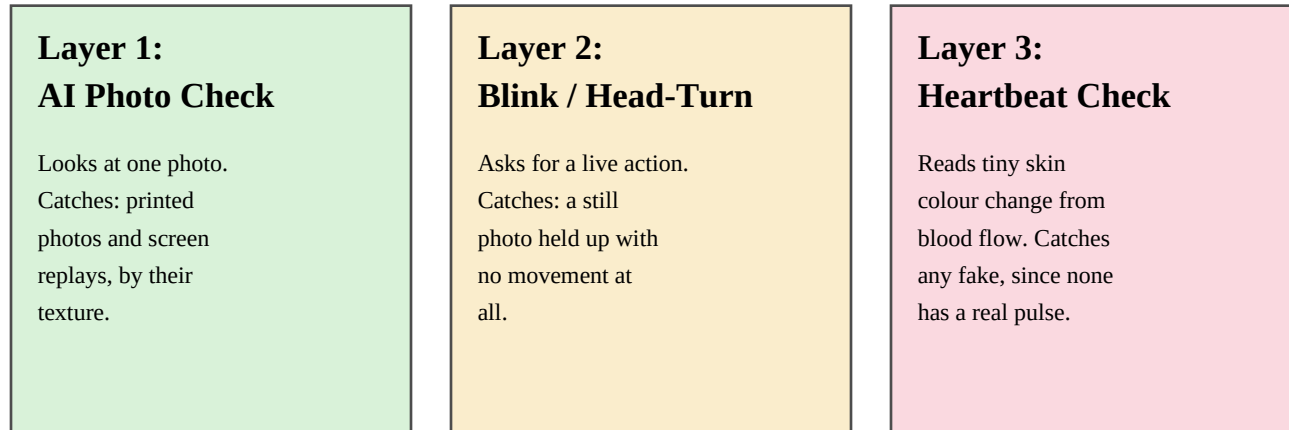
What we were trying to do: confirm that a real, live human being is in front of the camera right now, not a photo, a video playing on a screen, or a paused video frame.

What we actually built -- three separate, independent layers, so a weakness in one layer does not by itself let a fake through:

Layer 1: an already-trained AI model (built by others, not trained from scratch by us) that looks at a single photo and judges whether the skin and texture look like a real, living face or a printed/screen fake. Requires no user action.

Figure 2, next page, shows all three layers side by side.

Figure 2 -- The Three Liveness Layers



Each box catches something the other two might miss -- this is why all three run together instead of relying on just one.

Layer 2: an active challenge -- the system asks the user to do something live, such as blink or turn their head, and checks that a real, sustained movement actually happened, using the head-angle math from Phase 2 and a measurement of how open or closed the eyes are, frame by frame.

Layer 3: a heartbeat check called rPPG (remote photoplethysmography -- a method of detecting a heartbeat from video alone, with no sensor touching the skin). Every heartbeat pushes a tiny, invisible pulse of extra blood through the skin, very slightly changing its color in a rhythm matching the heartbeat. A printed photo or a screen replay has no real blood flow, so this signal simply does not exist in a fake.

Why three separate layers instead of one: research in this field shows that systems relying on only one method of checking liveness can be fooled, and we proved this ourselves directly rather than just citing outside research. When Layer 2 (the blink/head-turn challenge) was tested on its own against a recording of a real person's real blink, played back as a video, the system accepted it as a live blink, because the challenge only checks whether the correct movement pattern happened on screen, and a recording of the correct movement looks identical to the real thing. This is exactly why the system does not rely on Layer 2 by itself, and combines it with the other two layers instead.

Phase 4: Face Matching and Registration

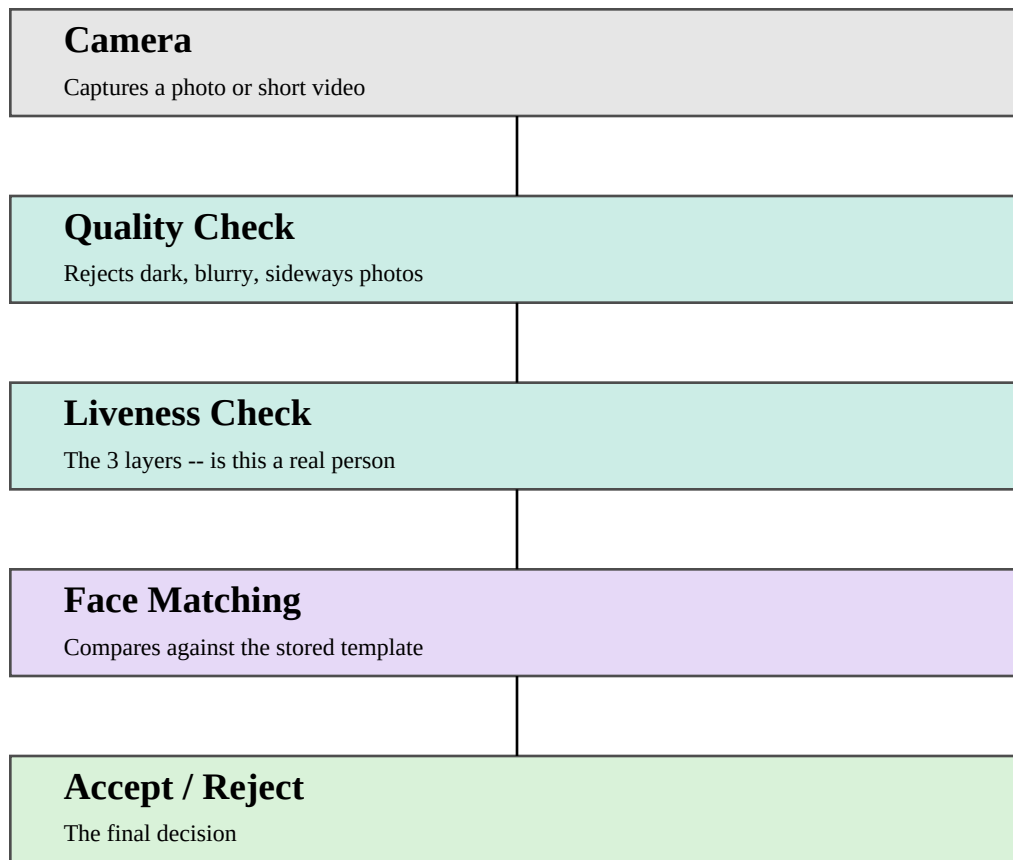
What we were trying to do: build the actual comparison that answers "does this face belong to the person who signed up," plus the process for signing someone up for the first time, called registration.

What we actually built: each face is converted into a list of numbers, called an embedding, using a well-established, pretrained model called ArcFace (accessed through a library called DeepFace). Two embeddings from the same person's face end up numerically close together; two embeddings from different people end up far apart. Comparing two embeddings with a similarity score is how matching works underneath.

The cutoff for "close enough to count as a match" is not guessed. It was calibrated using a real research dataset of face-pairs from different real people, plotting how the system performs at every possible cutoff and choosing a specific point deliberately, rather than picking a number that felt right.

Registration also includes a duplicate check: before storing a new person's face, the new face is compared against every already-registered face, and registration is refused if it looks like the same person is trying to register twice under a different name. This catches a simple but real kind of fraud at very low extra cost, since the same comparison logic already exists. Figure 3, next page, shows the complete pipeline built across Phases 2 to 4.

Figure 3 -- The Complete Pipeline



Phase 5: Mentor Feedback That Caused a Real Redesign

What we were trying to do: get outside, critical feedback on the design partway through the project, rather than only self-review.

The original design: quality checks worked as six separate, rigid pass-or-fail gates -- a photo either passed every single one of six hardcoded cutoffs, scattered across different parts of the code, or it was rejected outright, with no middle ground.

What the mentor flagged: this design is hard to explain to a client, since there is no single number to point to, and impossible for a client to adjust themselves. Not every user has a good camera or good lighting -- a rigid all-or-nothing gate that works fine in a bright test might reject many real users on an ordinary webcam in a dim room. Figure 4, next page, shows the redesign.

Figure 4 -- Before and After the Redesign

BEFORE

Brightness: pass / fail

Blur: pass / fail

Pose: pass / fail

Position: pass / fail

Single Face: pass / fail

Occlusion: pass / fail

AFTER

**1 Combined Quality Score
(0 to 100)**

+ one adjustable setting

Strict (85%+) -- high security

Balanced (70%+) -- everyday default

Lenient (50%+) -- older devices,
accessibility-first

The new design: every individual measurement now feeds into one combined quality score out of 100, instead of each acting as its own separate gate. A client can choose between three plain-language settings -- Strict, Balanced, or Lenient -- depending on how forgiving the system should be, shown as a literal dropdown menu in the interface, not a setting buried in code.

Why this matters: Strict suits a high-security setting where a slightly awkward photo should be rejected and re-taken. Lenient suits an accessibility-first setting or an older device, where being too strict would lock out real, legitimate users. Balanced is the sensible everyday default. This change turned a rigid, code-only decision into a real product setting a non-technical client can actually use and understand.

Phase 6: Security and Legal Readiness Review

What we were trying to do: before calling this ready for a real client, ask a harder question than "does it work" -- is it actually safe and legal to use with real people's faces.

What was missing, in plain terms: face data was being stored without encryption, meaning if the storage file were ever accessed by someone who should not have it, the raw face data inside would be directly readable.

There was no way to delete someone's data if they asked for it to be removed. This matters legally, not just technically -- most privacy laws covering biometric data (data about a person's body, like a fingerprint or a face) give people a legal right to have their data deleted on request, sometimes called the "right to be forgotten."

The backend system had no login protection of its own, meaning anyone who could reach it over the network could call it directly, with no check on who was asking.

What was added to fix each one: face data is now encrypted before it is ever saved to disk, using a standard, well-established encryption method, so the stored file is unreadable without the correct key.

Registration now requires an explicit, recorded consent step -- a checkbox the user must tick -- and is refused outright if consent was not given. A real delete function was added with two levels: a soft delete, which removes someone from all future matching immediately but keeps a recoverable record, and a hard delete, which permanently and irreversibly removes their data.

The backend now requires a security key on every request, plus a limit on how many requests can be made in a short period of time, to prevent abuse.

One more real problem was found and fixed during this same review: the very numbers used to judge "is this really the same person" had been tested using made-up, artificial comparison data standing in for a second person, not real photos of a real, different human being. This was caught and fixed by testing against real photo pairs of genuinely different people from a research dataset, instead of continuing to trust the artificial stand-in data.

Phase 7: The Final Honest Self-Check

What we were trying to do: right before considering this ready to demonstrate, stop trusting that each piece worked just because it had worked once, on its own, in isolation, and instead read through the whole system together, as it actually stood, end to end.

What this check found, stated plainly: several pieces that had each been built and tested successfully on their own had never actually been connected to each other in the running system. The heartbeat-detection layer from Phase 3 was built, calibrated, and tested, but was not wired into the live, real-time verification flow.

The duplicate-registration check from Phase 4 was built and tested, but the registration screen itself was not yet calling it, meaning a duplicate registration could have gone through undetected in the live product despite the check existing in the code.

One additional, separate bug was also found: the process that starts up the two halves of the system (the backend and the interface) could hang indefinitely under certain conditions, because a stream of output from one process was being collected but never actually read anywhere, filling up an internal buffer and causing the whole startup to freeze.

Being honest about the timing: this was found late, close to the point this project was going to be called finished, not early. It is reported here as found and now fixed, not hidden or quietly patched without mention. The duplicate-check was connected into the live registration screen, and the startup hang was fixed by making sure that output stream is properly read instead of left to fill up. The heartbeat-detection layer remains a deliberate, disclosed decision rather than a silent gap -- it needs several seconds of continuous, motion-free video to work, which does not fit the system's current moment-by-moment interaction style without a larger redesign, so it is documented as built, tested, and intentionally not yet part of the live flow, rather than presented as something it is not.

What Works Today

The following is a direct list of what is fully built, connected, and confirmed working in the live system right now -- not what was built in isolation at some point, only what is actually reachable by a real user today.

- Image quality checks run as one combined score with a Strict/Balanced/Lenient setting a client can choose.
- Passive liveness (Layer 1, the AI photo check) runs live on every attempt.
- Active liveness (Layer 2, blink/head-turn) runs live during both flows.
- Face matching runs against a calibrated, real-data-tested threshold.
- Registration, consent, and duplicate-check are fully connected and live.
- Data is encrypted at rest; both soft and hard delete work.
- The backend requires a security key and limits request rates.
- An automated test suite covering all of the above passes in full.

What Is Still Open or Needs More Work

This list is deliberately just as direct as the list of what works. Nothing here is softened, since a mentor and a client trust this report more when this section is specific, not vague.

- Screen-replay liveness gap: when a face is replayed from a laptop screen specifically, the passive AI check (Layer 1) misses it a large share of the time in testing -- confirmed as a real, repeatable weakness on a larger, expanded test sample, not a one-off fluke, and not yet fixed.
- The active-liveness bypass described in Phase 3 (a replayed recording of a real blink can pass the blink challenge) is a known, disclosed limitation, not something planned to be fixed without a larger design change, since it is exactly why the system does not rely on that layer alone.
- The heartbeat-detection layer (Layer 3) remains built and calibrated but not connected into the live, real-time flow, as explained in Phase 7.
- Fairness across different groups of people (for example, by gender, age group, or skin tone) has been measured on real test photos, and real, uneven gaps were found and written down, but no fix or rebalancing has been attempted yet. This should be disclosed to any client before wider use, not treated as resolved.
- A known, occasional camera-connection hiccup exists, where the live video feed can briefly and silently drop for twenty to thirty seconds before recovering on its own; this appears to be a characteristic of the underlying video-connection software itself, not a bug in this project's own code, and does not yet have a fix beyond restarting the connection.
- The demonstration video itself, called for in the original project brief, has not yet been recorded.
- One final confirmation pass, re-checking the most recent fixes together as one whole system, is still pending before this should be considered fully demo-ready.

Closing Summary

This project demonstrates a specific way of working, more than it demonstrates any single feature: build something, then honestly find out whether it actually works using real data and real tests, then fix what is genuinely broken rather than whatever is easiest to fix, and write down what is not yet solved instead of quietly hoping no one asks.

That approach showed up at every stage of this project. The head-angle bug in Phase 2 was found using real test footage, not assumed away. The three-layer liveness design in Phase 3 exists specifically because a weakness was proven in testing, not just read about. The mentor's feedback in Phase 5 led to an actual rebuilt design, not a cosmetic change. The security review in Phase 6 found and fixed real gaps, including a flawed test setup that was quietly giving an overly confident result. The final self-check in Phase 7 found real, disconnected pieces close to the finish line and said so plainly, rather than let a demonstration imply something was working when it was not yet confirmed.

The immediate next steps are short and known: confirm the fixes from Phase 7 hold up together as one system, then record the demonstration video, the one deliverable from the original brief still outstanding. Everything else in this report -- the working system, the honest list of what remains open, and the real numbers behind every claim -- is ready today.